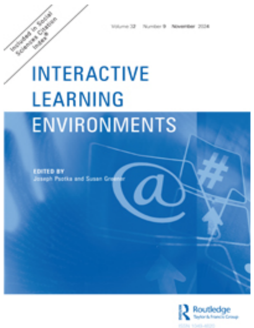

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Chatbot-based learning of logical fallacies in EFL writing: perceived effectiveness in improving target knowledge and learner motivation

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ABSTRACT

Chatbots have been increasingly applied for EFL education and demonstrated overall usefulness in improving knowledge and motivation, while this technology has yet to be used for learning logical fallacies (i.e. errors in reasoning) in EFL writing. However, knowledge of logical fallacies is essential, with which learners can avoid fallacies in EFL writing and have enhanced writing quality. To fill in the gap, this study investigated the perceived effectiveness of chatbots in developing knowledge of logical fallacy in EFL writing and enhancing learner motivation. Features of this learning method were also explored based on the comparison against website-based learning. Two groups of 15 Chinese EFL learners engaged in five-week autonomous, out-of-class, out-of-class learning of logical fallacies in EFL writing using a chatbot or a website. Semi-structured interviews, pre-post tests of fallacy knowledge and pre-post motivation questionnaires were conducted. The results showed that the chatbot was perceived as slightly less effective than the website in developing target knowledge but more effective in improving motivation. Compared to the website, chatbots were advantageous in high-quality human-computer interactions, study plan making, and high accessibility. Based on the research results, we discussed how this technology might influence fallacy learning based on the self-regulated learning framework.

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Chatbot; English writing; logical fallacy; second language writing; technology-enhanced language learning

Introduction

The chatbot is a digital tool for natural-language conversations and has been increasingly applied for educational purposes (Bailey et al., 2021). Compared to educational websites, chatbots share features in presenting multimedia-enhanced learning materials, affording human-computer interactivity, and supporting high learner autonomy (Nguyen & Sidorova, 2018), expected to become an excellent alternative to websites (Heo & Lee, 2018). In EFL education, researchers have identified the overall usefulness of chatbots in enhancing EFL knowledge and boosting motivation (Hsu et al., 2021; Tai & Chen, 2020), while this technology has rarely been used for learning logical fallacies in EFL writing. Logical fallacies refer to errors in reasoning (Saidi, 2020). In EFL writing, students frequently produce logical fallacies (Murray, 2012) that reduce the strength of claims and undermine the quality of entire writings (Song & Sparks, 2019). Learners' production of logical fallacies mainly resulted from their lack of knowledge of logical fallacies in EFL writing

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(El Khoiri & Widiati, 2017). Hence, to enhance EFL writing quality from the logical perspective, it is helpful to develop learners' knowledge of logical fallacies in EFL writing so they can identify and avoid this problem with writing (Lismay, 2020). Moreover, high motivation may enhance learning efficiency of logical fallacies (see Petri & Govern, 2012), further improving their fallacy knowledge and EFL writing proficiency. Hence, it appears beneficial to explore effective methods to improve students' knowledge of logical fallacies in EFL writing and motivation in fallacy learning. However, so far, research has remained few in this direction.

This article presents an exploratory study of applying a chatbot in learning logical fallacies frequently seen in EFL writing, based on the comparison with an educational website. Aiming to explore the perceived effectiveness of the chatbot in developing target knowledge of logical fallacies in EFL writing and improving motivation in fallacy learning, we conducted this study with the method of fallacy learning as the independent variable and learners' target knowledge of logical fallacy and motivation in learning fallacies as dependent variables. The advantageous, disadvantageous and mixedly-perceived features of this learning method were also explored. Based on the analysis of the results, we discussed various aspects from which chatbots may influence fallacy learning and provided implications for future research in this field.

It is noteworthy that the main learning objects of this study are knowledge of logical fallacies in EFL writing and motivation in learning fallacies. EFL writing is the context of the logical fallacies we focused on, not our main learning object.

Literature review

Logical fallacies in EFL writing

Logical fallacy refers to "an error in reasoning" (Johnson, 2009, p. 251), the situation when the reason could not adequately support the augment (Saidi, 2020). People produce logical fallacies in many situations, such as daily conversations and formal debates (Miller & Poston, 2020). In EFL writing, logical fallacies are about irrelevant claims and insufficient evidence (Murray, 2012). They lower the strength of arguments, undermine the logic in reasoning, and reduce the coherence of the presentation, negatively influencing the entire writing quality (Lismay, 2020; Murray, 2012). Hence, addressing logical fallacies is essential in improving EFL writing.

Unfortunately, most EFL learners, even the advanced ones, have difficulty addressing logical fallacies in writing (El Khoiri & Widiati, 2017). For example, El Khoiri and Widiati (2017) required 40 university students to write EFL argumentative essays and found that most students produced a surprisingly big number of fallacies in writing. Similarly, Lismay (2020) analysed EFL essays written by 28 university students, while finding that logical fallacies flawed over 76% of them. These results indicated the students' severe problem of logical fallacies in EFL writing.

To improve EFL writing from the logical perspective, developing students' knowledge of logical fallacies in EFL writing is necessary that includes the fallacy meanings, fallacy terms, and the dialectical contexts of EFL writing where fallacies occur (Saidi, 2020; Song & Sparks, 2019). Research showed that students' production of logical fallacies in EFL writing largely resulted from their lack of fallacy knowledge (El Khoiri & Widiati, 2017). Evidence was found by Song et al. (2020). They required 472 students to identify fallacies in argumentative writings and found that their students could not perform the task well without sufficient knowledge of logical fallacies. El Khoiri and Widiati (2017) argued that without knowledge of logical fallacies in EFL writing, students tended to have shallow awareness of reasoning. Under this circumstance, students may not even consider their writings from a logical perspective, let alone address the problem of logical fallacies. Thus, the researchers recommended helping students foster awareness of reasoning in EFL writing by developing their knowledge of logical fallacies from the aspects of meanings, terms, and dialectical contexts in EFL writing.

Chatbot-based learning and its comparison with website-based learning

A chatbot is a computer programme based on natural language processing and rule-based techniques that can interact with humans in text or voice (Ouatu & Gifu, 2021). For educational purposes, this technology is conducive to self-regulated learning (Maldonado-Mahauad et al., 2022; Pengel et al., 2021), which refers to learners' proactive management of academic tasks from four aspects: metacognitive, cognitive, emotional and behavioural (Dent & Koenka, 2016; Zhang & Zou, 2022). By dividing learning into "a series of micro-learning units and tasks" that learners can order, select and adapt according to individual needs and preferences (Yin et al., 2021, p. 169) and encouraging learners to apply self-regulated learning strategies, such as self-reflection and self-evaluation (Tandy et al., 2017; Tegos et al., 2019, november), chatbots can result in learners' active and constructive engagement in academic tasks (Ruan et al., 2020), leading to their high efficiency (Hsieh, 2011), strong satisfaction (Hsieh, 2011) and low cognitive loads (Yin et al., 2021).

In recent years, chatbots have been increasingly used and demonstrated usefulness in improving knowledge and learner motivation from different perspectives (Bailey et al., 2021; El Shazly, 2021). Based on the self-regulated learning framework, chatbots may influence learning from four aspects. From the metacognitive aspect, chatbots allow learners with high autonomy (Tandy et al., 2017) and substantial control over learning (Yin et al., 2021), enabling them to arrange learning tasks, schedules and materials based on individual needs and pace (Yin et al., 2021) and thereby achieve enhanced learning efficiency and motivation (Hsieh, 2011; Yin et al., 2021). From the cognitive aspect, chatbots could repeatedly present a huge amount of multimedia-enhanced learning materials, academic tasks and instructional support (Bailey et al., 2021), thereby allowing rich opportunities for learning (Ruan et al., 2019a) and practice (El Shazly, 2021). From the behavioural aspect, chatbot-based learning was featured by high accessibility and flexibility (Ouatu & Gifu, 2021), enabling learners to engage in learning tasks, retrieve learning materials, and receive feedback and recommendations whenever and wherever they like (Huang et al., 2022). From the emotional aspect, chatbots afford adequate, vivid human-computer interactivity in natural language (Ruan et al., 2019a) and thereby create social connections with learners as tutors and learning companions (Pérez et al., 2020). In this way, chatbots may increase learners' feelings of enjoyment (Tai & Chen, 2020) and comfortableness (Ruan et al., 2019b), conducive to their knowledge development.

Due to their usefulness, chatbots have been expected to become a powerful alternative to websites (Heo & Lee, 2018) that is the web-based digital tools widely used and overall effective for education. Researchers have compared chatbots with websites and identified several similarities. Nguyen and Sidorova (2018) contended that both websites and chatbots could repeatedly present multimedia-enhanced information, while websites tended to present information in a more complex and organised way than chatbots due to the larger interfaces of websites. Additionally, both websites and chatbots afford human-computer interactivity (Nguyen & Sidorova, 2018). However, chatbot-based interactivity was overall straightforward, based on natural language, while website-based interactivity was relatively complex, requiring users to "follo[w] a sequence of steps and interact[t] with different website components in order to transform their goals into commands" (p. 2).

Chatbot-based learning for EFL education

Researchers have applied chatbots to developing various aspects of EFL knowledge and skills, including vocabulary (Ruan et al., 2019a), speaking (Fryer et al., 2017), grammar (Nghi et al., 2019), and reading (Ruan et al., 2019b). The overall effectiveness of this technology has been identified in EFL education in improving target knowledge and learner motivation (El Shazly, 2021). For example, Nghi et al. (2019) required 100 university students to develop EFL grammar knowledge using a chatbot that could repeatedly present multimedia-enhanced learning materials. The results showed that chatbot-based learning significantly enhanced the participants' grammar knowledge and motivation. In another study, Hsu et al. (2021) required 24 university students to develop

EFL speaking skills using a mobile chatbot that could present EFL live broadcasts, provide speaking exercises, evaluate learners' spoken output and provide immediate feedback. The results showed that students obtained significant progress in EFL speaking after four-month chatbot-based learning.

The overall effectiveness of chatbots in EFL education may be consistent in learning logical fallacies in EFL writing. As indicated by previous literature on chatbot-based EFL education (e.g. Bailey et al., 2021; El Shazly, 2021; Ruan et al., 2019b; Tandy et al., 2017), chatbots may help develop target knowledge of logical fallacies in EFL writing by providing multimedia-enhanced learning materials, various types of exercises, feedback, guiding questions, and recommendations without temporal or spatial constraint. This technology may also improve motivation in learning logical fallacies in EFL writing by supporting rich and straightforward human-computer interactivity, affording high flexibility and accessibility of learning, encouraging proactive regulation of learning, and presenting various entertaining elements.

Research gaps

We identified two research gaps in EFL writing development and chatbot-based learning. Firstly, educational chatbots have seldom been applied for learning logical fallacies in EFL writing, an essential aspect of EFL writing knowledge and a critical measure of writing quality (El Khoiri & Widiati, 2017). However, being overall effective in improving EFL knowledge and motivation in EFL learning (see Ruan et al., 2019a, 2019b), chatbots demonstrate great potential in this understudied field. Hence, it appears valuable to investigate chatbot-based learning of logical fallacies in EFL writing so as to extend the educational arenas for chatbot application and explore the approaches to improving EFL writing from the logical perspective.

Furthermore, few studies have compared chatbots with websites for EFL learning. However, these two technology tools are highly comparable in educational contexts, sharing features in presenting multimedia-enhanced information and supporting human-computer interactivity (Nguyen & Sidorova, 2018). Some researchers even considered chatbots a powerful alternative to websites (Heo & Lee, 2018). Comparing these two technology tools can result in a more in-depth understanding of educational chatbots and provide implications for researchers and practitioners in selecting and applying technology for EFL educational purposes (Huang et al., 2022). Hence, it appears meaningful to investigate the perceived effectiveness and features of chatbot-based EFL learning based on the comparison with the website-based one.

Research questions

To fill the gaps, this article presents an exploratory study of chatbot-based learning of logical fallacies in EFL writing based on the comparison with website-based learning. This study focused on the effects of the learning method on improving target knowledge of logical fallacies in EFL writing and motivation in learning fallacies. The advantageous, disadvantageous, and mixedly-perceived features of chatbot-based learning compared to website-based learning were also explored. Three questions guided this study:

> RQ1: Compared to the website, is the chatbot perceived as effective in developing target knowledge of logical fallacies in EFL writing?

> RQ2: Compared to the website, is the chatbot perceived as effective in increasing motivation in learning logical fallacies in EFL writing?

> RQ3: Compared to website-based learning, how is the chatbot advantageous, disadvantageous and mixedly-perceived in learning logical fallacies in EFL writing?

We conducted semi-structured interviews to address the three research questions. In addition, we conducted pre-post tests of the target knowledge of logical fallacy to triangulate the interview data concerning RQ1 and pre-post questionnaires of learner motivation to triangulate the interview data concerning RQ2. Under the Result section, sub-section "Perceived effectiveness of chatbot-based learning in developing target knowledge of logical fallacies in EFL writing" presents the results concerning RQ1; Sub-section "Perceived effectiveness of chatbot-based learning in increasing motivation in learning logical fallacies in EFL writing" presents the results concerning RQ2; Sub-section "Advantageous and mixedly-perceived features of chatbot-based learning of logical fallacies in EFL writing" presents the results concerning RQ3. Under the Discussion section, we analysed the research results and explored how chatbots might influence fallacy learning based on the self-regulated learning framework, as presented in sub-section "How chatbots may influence fallacy learning: From the perspective of self-regulated learning". We also suggested directions for future studies in this field, as presented in sub-section "Implications for researchers".

Chatbot-based learning of logical fallacies in EFL writing

The authors created a five-week chatbot-based learning programme to teach the term-meaning of logical fallacies and dialectic contexts of fallacies frequently seen in EFL writing. Twenty-two target fallacies were identified from the literature review (e.g. Bassham et al., 2010; El Khoiri & Widiati, 2017; Lismay, 2020; Murray, 2012; Song et al., 2020). Appendix A presents the syllabus. The programme consists of two main components: learning tasks and the chatbot artefact.

Learning tasks

With reference to the recommendations in the previous studies (e.g. El Khoiri & Widiati, 2017; Saidi, 2020), we integrated eight types of learning materials into chatbot-based learning to help develop target knowledge of logical fallacies in EFL writing. (a) Instructions presented in texts and images. The multimedia-enhanced instructions present the taxonomy and explanations of the 22 target logical fallacies, from which learners may learn about the term-meaning of the target fallacies. (b) Example sentences of EFL writing. The example sentences are categorised according to the logical fallacies occurring therein, from which learners may obtain a vivid understanding of the dialectic contexts of the target fallacies in EFL writing. (c) Supplementary videos from Youtube (www.youtube.com). All videos are within 2 min, illustrating the meanings, forms and contexts of the target fallacies in a vivid, conversational manner, helping learners understand and memorise the target knowledge efficiently. (d) Glossary of logical fallacies in EFL writing. A glossary presents the term-meaning of the 22 target logical fallacies in an organised way. Using the glossary, the learners may efficiently conduct previews and reviews of the target fallacy knowledge. (e) Multiple choice and True/False exercises, two types of exercises that are frequently used in fallacy education (e.g. in Bassham et al., 2010). By doing these exercises, students can reinforce their memory of the target knowledge, differentiate similar logical fallacies, and enhance their understanding of some tricky fallacies. (f) Quizlet-based flashcard exercises (https://quizlet.com/hk/662674485/logical-fallacy-flash-cards/). The front side of the cards presents EFL sentences where some logical fallacies occur, and the opposite side presents the exact type of fallacies occurring in the sentences. By doing the flashcard exercises, learners may check and reinforce their understanding of the target fallacies and enhance their understanding of the fallacy contexts of EFL writing. (g) Self-evaluation checklist. The chatbot presents a series of guiding questions, aiming to lead learners to reflect on their process and performance in fallacy learning. The learning materials were composed with reference to Bassham et al. (2010), El Khoiri and Widiati (2017), Murray (2012), Softschools (https://www.softschools.com/examples/fallacies/) and the website of the Department of Philosophy, Texas State University (https://www.txstate.edu/philosophy/resources/fallacy-definitions/).

Educational chatbot

The first author created the educational chatbot using ManyChat (https://manychat.com/). The chatbot was based on Facebook Messenger, a mobile social media application widely used by young people and frequently applied in chatbot-based learning (Nghi et al., 2019).

Eight chatbot functions were applied to assist the learning of logical fallacy in EFL writing that were frequently applied in previous chatbots for EFL education (e.g. Bailey et al., 2021; Chen et al., 2020 ; Ruan et al., 2019a, 2019b; Tai & Chen, 2020). They are: (a) presenting multimedia-enhanced learning materials (i.e. instructions, examples, and glossary); (b) presenting hyperlinks to external learning materials and exercises (i.e. videos, Quizlet-based flashcard exercises); (c) asking content-related questions (e.g. "Do you understand the meaning of 'Red Herring'?"); (d) presenting exercises (i.e. multiple choice exercises and true/false exercises); (e) evaluating learners' performance and giving immediate feedback (e.g. "Correct!", "It is not correct."); (f) asking guiding questions (e.g. "Can you figure out the type of logical fallacy in this sentence?"); (g) providing encouragement and praise (e.g. "You are doing very well!"); (h) offering giving recommendations (e.g. "You may refer to the example sentences for enhanced understanding."). Various conversational elements featured the chatbot's messages to increase the authenticity and fun of the human–computer interactivity, as recommended by Pereira (2016), including emojis and casual conversational words (e.g. "SAVAGE!"). Users could interact with the chatbot by selecting options and typing down words. Figure 1 display some sample interfaces of this chatbot.

Methods

Participants

This study involved 30 Chinese English learners (25 females and five males) based on voluntary response sampling. They were students at a university in Hong Kong, aged between 21 and 34. All participants had over ten-year experiences of EFL writing and overall upper-intermediate English proficiencies. No participant had previously received formal or informal learning related to logical fallacies in EFL writing. They all had smartphones and computers for browsing websites and using apps. All participants had used educational websites before. All participants had used chatbots before but never for educational purposes.

Experimental procedures

All participants were required to learn the target knowledge of logical fallacies in EFL writing for five weeks based on the same learning materials. During the five weeks, the participants learned fallacies out of class with complete autonomy, no interaction with other participants, and no intervention from the researchers other than one study reminder through email and one private online conversation concerning their learning progress per week. We used an online random team generator (https://www.randomlists.com/team-generator) to randomly and equally divided the participants into two groups based on the two-level independent variable of this study: the method of learning logical fallacies. One was the chatbot-assisted learning group, in which the participants engaged in the five-week learning using the researcher-developed chatbot; The other was the website-based learning group, in which the participants engaged in the learning using a researcher-developed website (https://s1139421.wixsite.com/logicalfallacy). According to the user behaviours recorded by the chatbot and website managements, the chatbot-based learning group participants, on average, had used the chatbot 16.80 times (SD = 9.22) during the five weeks.

The participants were required to complete pre-post tests of target fallacy knowledge and pre-post motivation questionnaires. We also invited the participants to semi-structured interviews to report their experiences and perceptions of technology-enhanced learning after the learning period.

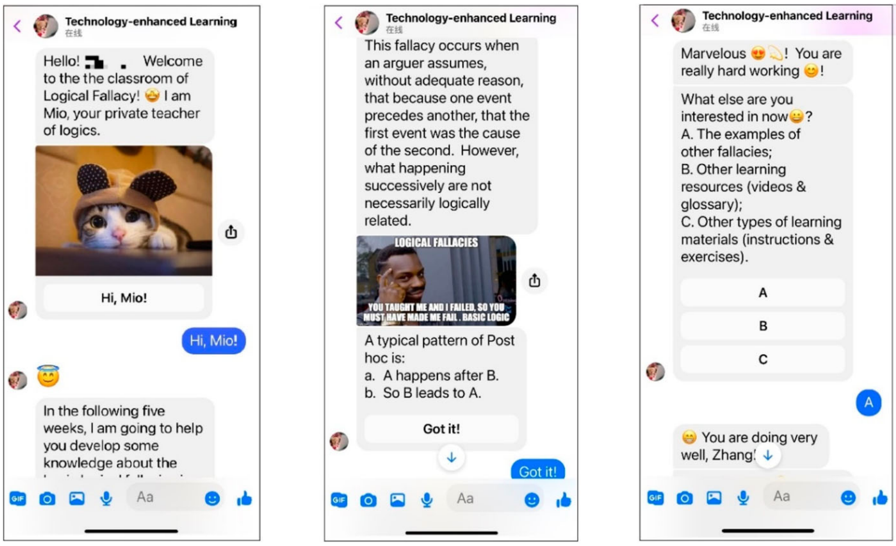

Figure 1. Sample interfaces of the educational chatbot.

Data collection

We conducted three data collection methods. The first method is semi-structured interviews to address RQ1, 2 and 3. The second is pre-post tests of the target knowledge of logical fallacy in EFL writing to triangulate the interview results concerning RQ1. The third method is pre-post questionnaires of learner motivation in learning logical fallacies to triangulate the interview results concerning RQ2. The following sub-sections present the details of the three methods.

Semi-structured interviews

We conducted semi-structured interviews following the procedures adapted from DiCicco-Bloom and Crabtree (2006) to collect data about learners' perceived effectiveness and features of the learning methods in developing target fallacy knowledge and enhancing motivation in fallacy learning. Three categories of questions were used corresponding to the three research questions: (a) the perceived effectiveness of the learning methods in developing the target knowledge of logical fallacy in EFL writing (e.g. "Do you think the chatbot-assisted learning/ website-based learning of logical fallacies effective for developing target fallacy knowledge in EFL writing? Why?"); (b) the perceived effectiveness of the learning methods in enhancing motivation in learning fallacies (e.g. "Did the chatbot-assisted learning/ website-based learning of logical fallacies influence your motivation in learning fallacies in EFL writing? Why?"); (c) the advantageous, disadvantageous and mixedly-perceived features of the learning methods (e.g. "What's helpful/unhelpful in your chatbot/website-based learning of logical fallacies in EFL writing? Why?"). The interviews were conducted for around 40 min conversationally in Chinese, the participants' L1, so they felt free to express their opinions, following Zhang et al. (2021).

We conducted mock semi-structured interviews before the study with five adults with similar demographic backgrounds to the participants in the primary study. The results of the mock interviews indicated the reliability and validity of the interview questions and the interview methods in collecting the target information.

Pre-post tests of the target logical fallacy knowledge

We conducted pre-post tests of target knowledge of logical fallacies in EFL writing to triangulate the interview results concerning learners' perceived effectiveness of the learning methods in developing target knowledge. The test is based on 22 EFL sentences developed with reference to Bassham et al. (2010) and Murray (2012), each flawed by one of the 22 target types of logical fallacies. The test system was adapted from the CBAL Analysing Arguments assessment, consisting of two parts: (a) the recognition of the particular type of logical fallacy in each of the 22 EFL sentences, scored from 0 (when the participant could not recognise the particular type of logical fallacy) to 2 (when the participant recognises the particular type of logical fallacy and explain the fallacy meaning without mistake); (b) the matching between the meanings and terms of logical fallacies, scored from 0 (when the participant fails to match the fallacy term with the corresponding meaning) to 1 (when the participant correctly matches the fallacy term with the corresponding meaning). Hence, the test's total score is 66, 3 points at maximum for each of the 22 target fallacies. Appendix B presents the test details.

We conducted mock tests before the study with five adults with similar demographic backgrounds to the participants in the primary study. The results indicated the reliability and validity of the test in measuring the target knowledge of logical fallacies in EFL writing.

To ensure internal reliability, the contents of the pre-post tests were the same, despite the differences in the order of example sentences. To ensure external reliability, the test was blinded scored by the first and the second authors.

Pre-post questionnaires of learner motivation

We conducted pre-post questionnaires of motivation in learning fallacies to triangulate the interview results concerning learners' perceived effectiveness of the learning methods in increasing learner motivation. The questionnaire was based on a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). The questionnaire was adapted from Wu's (2018) learner motivation scale, consisting of 24 questions (e.g. "The fallacy learning activity met my needs."). This scale was adopted because it was used to test university students' motivation in technology-enhanced EFL learning and demonstrated high consistency and reliability of all sub-scales (Cronbach's $\alpha > .95$). Appendix C presents the questionnaire details.

Data analysis

To address the three research questions, the authors audio-recorded and transcribed the interviews together and then analysed the transcripts independently through a five-step method adapted from Braun and Clarke (2006). First, the authors read the transcripts repeatedly to get a general idea. Secondly, the authors summarised the participants' opinions and categorised them according to their involvement with learners' perceived effectiveness and features of chatbot-based/website-based learning in developing target fallacy knowledge and enhancing motivation in fallacy learning, corresponding to the three research questions. Thirdly, the summarised opinions were further categorised according to the features of the two learning methods mentioned by the participants. Fourthly, valid opinions supported by explicit reasons and examples were selected, with their frequencies of getting mentioned by the participants calculated. Representative transcripts were also quoted. Finally, the authors compared the classified opinions of the participants in the experimental and website-based learning groups to find the features of the two learning methods that the participants perceived differently. The authors compared their independent coding results and reached a satisfactory agreement with the remaining differences resolved via discussion. All participants received copies of the coding results for a second member check. Finally, the authors collaboratively translated the categorised opinions and quotations into English.

We conducted descriptive analyses of (a) students' scores in the pre-post tests of target knowledge of logical fallacies to triangulate the interview results concerning RQ1; (b) students' scores in the pre-post motivation questionnaires to triangulate the interview results concerning RQ2. The means and standard deviations of the scores were described in the analyses.

Results

This section presents the results of this study concerning the perceived effectiveness of the chatbot-based learning method in developing target knowledge of logical fallacies and increasing motivation in learning logical fallacies, based on the comparison to the website-based learning method. Interview results concerning the advantageous, disadvantageous and mixedly-perceived features of chatbot-based learning are also reported.

Perceived effectiveness of chatbot-based learning in developing target knowledge of logical fallacies in EFL writing

The interview results showed that chatbot-based learning was perceived as slightly less effective than website-based learning in developing target knowledge of logical fallacies, as shown in Figure 2. Ten chatbot-based learning group participants (67%) agreed or strongly agreed with the chatbot's effectiveness in developing target fallacy knowledge, and 12 website-based learning group participants (80%) agreed or strongly agreed with the website's effectiveness. Additionally, more participants disagreed or strongly disagreed with the chatbot's effectiveness in target knowledge development (3 participants, 20%) than those on the website (1 participant, 7%).

The descriptive statistics of test results, as shown in Table 1, triangulated the interview results. On average, the chatbot-based learning group obtained 10.57 (SD = 5.56) in the pre-test and 28.29 (SD = 12.09) in the post-test; the website-based learning group obtained 11.66 (SD = 7.53) in the pre-test and 31.03 (SD = 10.25) in the post-test. The results indicated slightly less effectiveness of the chatbot than the website in developing target knowledge of logical fallacies.

Perceived effectiveness of chatbot-based learning in increasing motivation in learning logical fallacies in EFL writing

The interview results showed more perceived effectiveness of the chatbot than the website in improving motivation in fallacy learning. As shown in Figure 3, eight participants agreed or strongly agreed with the effectiveness of chatbot-assisted learning in enhancing their motivation in learning fallacies (53%), and five participants agreed or strongly agreed with the effectiveness of website-assisted learning in enhancing their motivation (33%).

The questionnaire results, as presented in Table 2, triangulated the interview results. On average, the chatbot-based learning group obtained 76.86 (SD = 13.81) in the pre-questionnaire and 83.29 (SD = 13.30) in the post-questionnaire; the website-based learning group obtained 78.48 (SD = 10.55) in the pre-questionnaire and 83.22 (SD = 10.95) in the post- post-questionnaire. The results

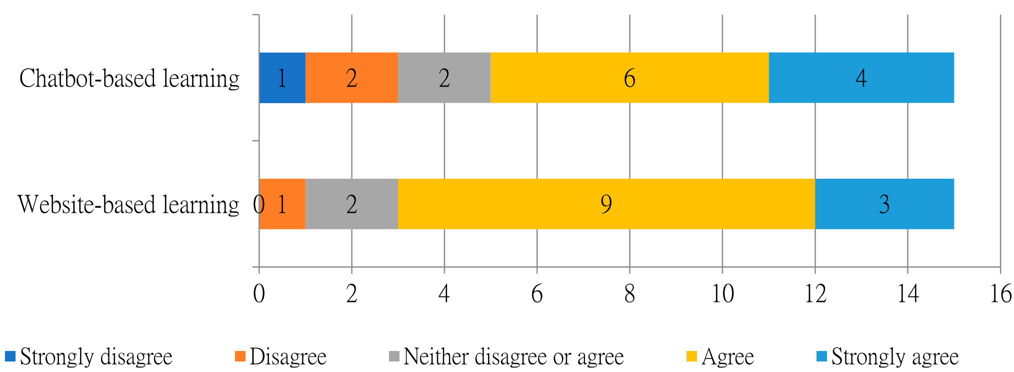

Figure 2. Frequencies of various degrees of participants' agreement on the effectiveness of chatbot-based/ website-based learning in developing target knowledge of logical fallacies in EFL writing.

Table 1. Descriptive statistics of pre-and-post tests of target knowledge of logical fallacies in EFL writing.

	Pre-test		Post-test	
	M	*SD*	*M*	*SD*
Chatbot-based learning group ($N = 15$)	10.57	5.56	28.29	12.09
Website-based learning group ($N = 15$)	11.66	7.53	31.03	10.25

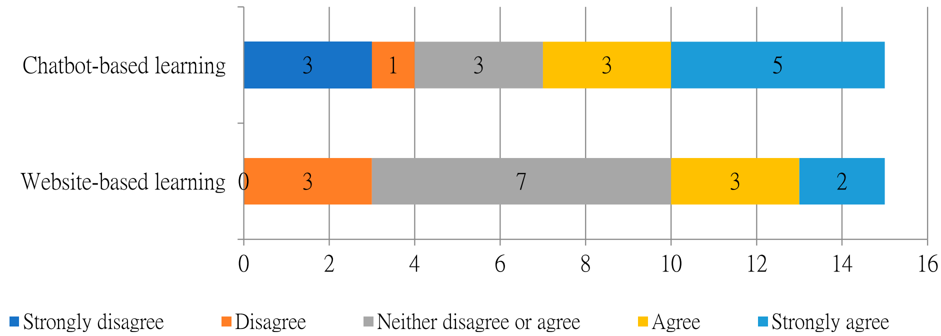

Figure 3. Frequencies of various degrees of participants' agreement on the effectiveness of chatbot-based/ website-based learning in improving motivation in fallacy learning.

suggested the more positive effects of chatbot-based learning than website-based learning on motivation in learning logical fallacies.

Advantageous and mixedly-perceived features of chatbot-based learning of logical fallacies in EFL writing

As shown in Figure 4, six features of chatbot-based learning of logical fallacies were mentioned most frequently in the interviews: presentation of learning materials (15 times), conversational elements (14 times), human-computer interactions (13 times), learner concentration and perseverance (ten times), supportiveness for study plan making (for four times) and accessibility (four times). All these features were also mentioned by the participants in the website-based learning group, except conversational elements (see Figure 5).

By comparing learners' perceptions of chatbot-based and website-based learning of logical fallacies, we identified three advantageous features, three mixedly-perceived features and no disadvantageous feature of chatbot-based learning compared to website-based one. Details of the advantageous and mixedly-perceived features are presented in the following subsections.

Advantageous features of chatbot-based learning of logical fallacies in EFL writing

The interview results revealed three main advantages of the chatbot over the website in learning logical fallacies in EFL writing. The first advantage concerned human-computer interactions, mentioned by all participants in the chatbot-based learning group (100%). The chatbot asked learners about their learning progress from time to time and provided encouragement and suggestions according to learners' responses, of which most participants perceived a strong sense of "freshness"

Table 2. Descriptive statistics of pre-and-post questionnaires concerning motivation in learning fallacies.

	Pre-questionnaire		Post- questionnaire	
	M	*SD*	*M*	*SD*
Chatbot-based learning group ($N = 15$)	76.86	13.81	83.29	13.30
Website-based learning group ($N = 15$)	78.48	10.55	83.22	10.95

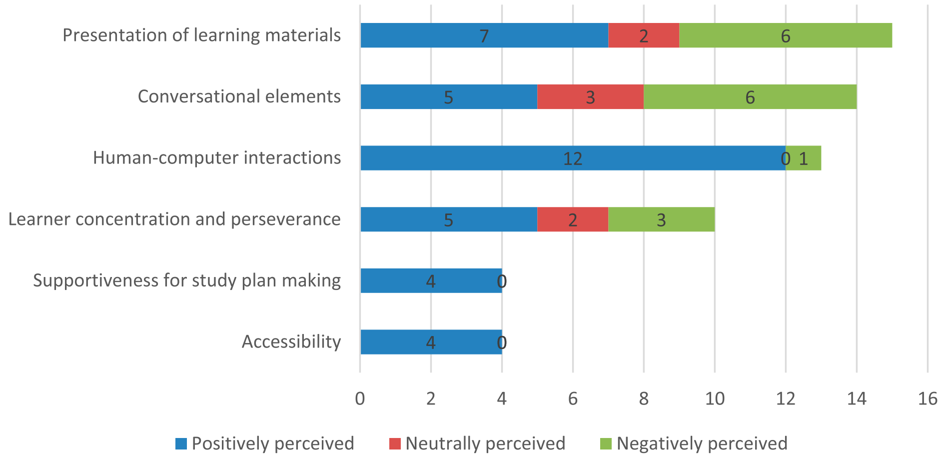

Figure 4. Frequency of different features of chatbot-based learning of logical fallacies in EFL writing that were positively, negatively, and neutrally perceived by learners.

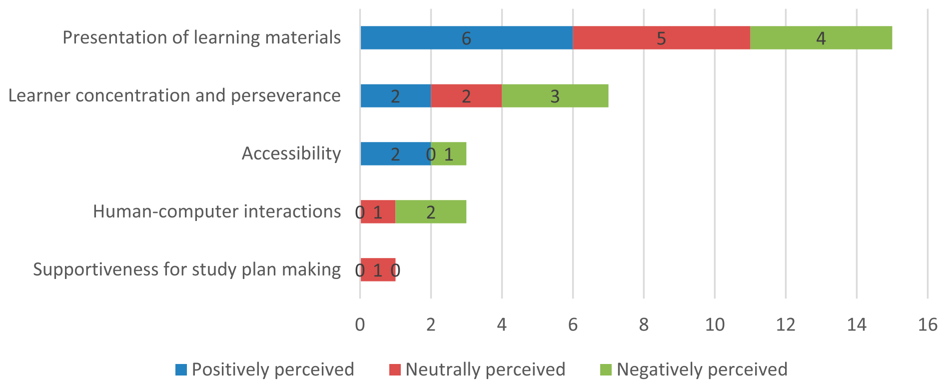

Figure 5. Frequency of different features of website-based learning of logical fallacies in EFL writing that were positively, negatively, and neutrally perceived by learners.

and "fun". Furthermore, by interacting with learners, the chatbot played the role of a supportive and patient friend in learning, which reduced learners' anxiety, loneliness and frustration, as commented by many participants. Another benefit of interactivity regards self-evaluation. When asked about their learning progress by human teachers and peers, learners tended to feel anxious, stressed, and fear of being judged. Such negative emotions may lead to their avoidance of genuine self-reflection, as some interviewees admitted. However, when it was the chatbot that asked about their learning progress, learners held no such negative emotions because "[t]he chatbot ha[d] no feeling, it would never judge learners, and it would always be patient", quoted from one participant in the chatbot-based learning group. Thus, chatbot-based learners tended to feel comfortable reviewing their learning process and assessing their academic performance, thereby conducting high-quality self-evaluation. Additionally, as reported by interviewees, the immediate feedback provided by the chatbot via interactivity assisted their knowledge reinforcement in exercises. "Using the chatbot, I could check my understanding of fallacies immediately after I completed the exercises. In this way, I can memorise the knowledge well". Quoted from one participant in the chatbot-based learning group. In contrast, some participants in the website-based learning group admitted that they gave up exercises due to the lack of immediate feedback.

Furthermore, chatbot-based learning was advantageous for supporting study plan creation, as mentioned by four participants in the chatbot-based learning group (27%). According to these participants, the chatbot clearly explained the purposes of various learning resources and the relationships between different learning tasks, conducive to learners' proactive arrangement of the types, amount, and order of learning tasks according to individual needs. In website-based learning, however, although the toolbar and systematic collection of webpages also illustrated the purposes of different learning resources, the relationships among the recourse were not specified.

The third advantage of the chatbot regards accessibility, mentioned by four participants in the experiment group (27%). They argued that chatbot-based learning was integrated into a mobile social media application they frequently used. Once the learners used the application, they saw the chatbot, which increased their learning opportunities. In contrast, although the website was also available on mobile phones, the participants mentioned that they did not use mobile websites as frequently as the social media application. Therefore, they considered chatbot-based learning more accessible than website-based one.

Mixedly-perceived features of chatbot-based learning of logical fallacies in EFL writing

The participants had mixed perceptions of three features of chatbot-based learning of logical fallacies in EFL writing. One concerns the presentation of learning materials. Seven chatbot-based learning group members (47%) mentioned their appreciation of the "one-at-a-time" presentation of learning materials in chatbot-based learning: The chatbot provided one knowledge point at a time in several chat boxes and presented another knowledge point only after the learners reported their complete understanding of the previous one. In this way, the chatbot guided learners to sufficiently comprehend knowledge points and gradually connect them into a holistic system in their memory "like real teachers do in authentic EFL classrooms", according to the participants. Exercises were also presented in a "one-at-a-time" way in chatbot-based learning, i.e. no more exercise would be shown before the learners have completed the previous one and received feedback. According to the chatbot-based learning group members, the "one-at-a-time" presentation of exercises helped them focus on the question at hand. Moreover, the chatbot-based learning group participants argued that the learning materials delivered "one-at-a-time" appeared "easy to handle", reducing their stress and reluctance to learn and practice. In website-based learning, however, learning materials were delivered "all-at-once" on webpages. Four participants in the website-based learning group (27%) admitted that they found instructions and exercises delivered "all-at-once" as "daunting" and "demotivating". However, not all students appreciated the "one-at-a-time" presentation of learning materials. Six participants (40%) argued that the instructions presented "one-at-a-time" in different chat boxes appeared "unfocused" and "unsystematic" and reduced their review efficiency. In contrast, six participants (40%) in the website-based learning group expressed appreciation for the systematic and complete presentation of instructions on the website. Another problem resulting from the "one-at-a-time" presentation of learning materials in chatbot-based learning was that the learners could hardly view different knowledge points simultaneously. However, according to some participants, simultaneously viewing different knowledge points was necessary to compare and differentiate logical fallacies, which was essential for learning the knowledge. Thus, by delivering instructions discontinuously, the chatbot may negatively influence the learners' efficiency in reviewing fallacy knowledge and comparing various logical fallacies.

Another mixedly-perceived feature of chatbot-based learning of logical fallacies waningwas the conversational elements, i.e. emojis and casual conversational words. Five participants (33%) had expressed overall appreciation of these elements, regarding them as "fun" and motivating. However, six participants (40%) held different opinions, perceiving only slight and declining positive effects of these conversational elements. They argued that they had a strong awareness of the chatbot's nonhuman nature, so the chatbot's conversational elements could not affectively influence them. "The chatbot is just a computer programme. The praises and encouragements from the chatbot represent no real feeling, so it triggers no real feeling [of mine]". Quoted from one participant in the chatbot-based learning group. Moreover, learners obtained increasing familiarity with the chatbot over time, leading to a waning sense of freshness in the human-chatbot interaction and the declining effectiveness of the conversational elements. "[The conversational elements] are interesting to me at the beginning, but then I just got bored of them over time". Commented by one participant in the chatbot-based learning group.

The third mixedly-perceived feature of chatbot-based learning of logical fallacies concerns learner concentration and perseverance. Five participants in the chatbot-based learning group (33%) reported that they could engage in chatbot-based learning with high concentration for a long time because of their long-term habits of chatting on social media. In their daily chat with friends on social media, they tended to be "triggered" by the new message notifications, focus on the contents in chat boxes excitedly, and deeply engage in conversations for a long time. As an extension of this habit, the learners felt the same stimulation when receiving the chatbot's instructional message. They focused on learning resources in chat boxes with high concentration for a long time. However, three participants in the same group (20%) held contrary opinions. They argued that chatbot-based learning was based on mobile phones in which they constantly received various new information and got distracted from fallacy learning.

Discussions

How chatbots may influence fallacy learning: from the perspective of self-regulated learning

We analysed the research results concerning the perceived effectiveness, advantages and disadvantages of chatbot-based learning in developing target knowledge of logical fallacies in EFL writing and enhancing motivation in fallacy learning. The analysis was conducted based on the self-regulated learning framework, with foci on various factors in chatbot-based learning of logical fallacies, the dynamics between these factors and the conditions of the varying dynamics. Based on the analyses, we identified how chatbots might influence fallacy learning from emotional, cognitive, metacognitive and behavioural aspects.

From the emotional aspect, chatbots may influence fallacy learning with human factors and nonhuman nature. Chatbots afford various human factors, such as human-chatbot interactivity and conversational elements. According to the interviews, these human factors enable chatbots to imitate human conversational behaviours and develop social connections with learners, enhancing learners' affective states, motivation and efficiency in fallacy learning. This finding is consistent with Huang et al. (2022) and Hsieh (2011). Our interviewees also reported the learners' sense of novelty and freshness for the human factors of chatbots, which might have further enhanced their motivation in fallacy learning, especially at the initial stage. This pheromone was also observed by Fryer et al. (2017). Nonetheless, chatbots are nonhuman, despite the effects of various human factors. The interviews showed that participants had a growing awareness of the chatbot's nonhuman nature, along with their increasing familiarity with this technology. Such awareness, as shown in the interview results, can free learners from interpersonal stress and anxiety, increasing learner motivation in fallacy learning and willingness to conduct self-evaluation. On the other hand, the interview results also indicated that the strong awareness of the nonhuman nature of chatbots could reduce the positive effects of human factors on learner motivation and efficiency in fallacy learning. This finding is consistent with Huang et al.'s (2019) observation. In this way, chatbots' human factors and nonhuman nature may have mixed effects on fallacy learning from the emotional aspect.

From the cognitive aspect, chatbots may influence fallacy learning with the "one-at-a-time" presentation of learning materials and immediate feedback. Regarding knowledge acquisition, the interview results indicated that the "one-at-a-time" presentation of learning materials in chatbot-assisted learning led to sufficient processing of new information and the construction of mental connections between different knowledge points, conducive to developing target knowledge of logical fallacies. This finding echoed Nguyen and Sidorova's (2018) analysis of the advantages of chatbot-based information presentation over website-based one. Similarly, in knowledge consolidation, the "one-at-a-time" presentation of exercises can improve learner concentration and efficiency in exercises, as shown in the interview results. The "one-at-a-time" presentation of learning materials may also enhance motivation in learning and practising, since, according to the interview results, learners tended to consider the instructions and exercises "easy-to-handle" when presented in the "one-at-a-time" way. Moreover, chatbots afforded immediate feedback on learner performance in exercises, which, according to our participants, was helpful for knowledge consolidation. This finding is aligned with the theory of the negative suggestion effect, which contends that in-time feedback in exercises helps learners memorise the correct information with minimal risk of reinforcing the wrong information (Roediger & Butler, 2011). Hence, the immediate feedback in chatbot-based exercises may also enhance the development of fallacy knowledge. Regarding knowledge internalisation and elaboration, the "one-at-a-time" presentation of learning materials may reduce learner concentration and efficiency in reviewing and comparing knowledge points. However, according to the interview results, reviewing and comparing knowledge points was essential for internalising and comprehending fallacy knowledge. The result is also triangulated by the test scores, which showed the chatbot-based learning group's slightly less achievement of fallacy knowledge than the website-based learning group. Thus, chatbots may enhance learners' acquisition and consolidation of target knowledge of logical fallacies in EFL writing but reduce their efficiency in knowledge internalisation and elaboration due to the "one-at-a-time" presentation of learning materials and immediate feedback.

From the metacognitive aspect, chatbots may influence fallacy learning by scaffolding the application of planning and self-evaluation. As for planning, the interview results showed that chatbots could scaffold study planning by providing recommendations and explicit information about the specific purposes of different learning tasks and the relationships among these tasks. As for self-evaluation, chatbots can provide guiding questions about learners' progress and achievement to trigger and scaffold their self-evaluation. Due to the nonhuman nature of chatbots, learners tend to be self-honest and stress-free in reviewing and quality assessing, thus having high frequency and efficiency of self-evaluation in chatbot-based learning, as shown in the interview results. Our participants reported their overall positive perceptions of planning and self-evaluation in chatbot-based learning of logical fallacies in EFL writing, aligned with the theories of self-regulated learning (see Zimmerman & Moylan, 2009). According to the theories, planning and self-evaluation help learners control and regulate their entire learning process, increasing their motivation to learn and expediting their achievement of learning goals (Zimmerman & Moylan, 2009). Hence, chatbots may enhance fallacy learning from a metacognitive aspect by supporting learner application of planning and self-evaluation.

From the behavioural aspect, chatbots may influence fallacy learning through their integration with mobile social media applications. As revealed in the interviews, chatbots integrated with mobile social media applications were likely to distract learners from academic tasks because of the various other applications and notifications on mobile phones, leading to reduced learning efficiency. Nonetheless, integrating chatbots with mobile social media applications may result in easy access to chatbot-based learning of logical fallacies, as reported by our participants, providing learners with extensive exposure to learning and practising the target knowledge. Moreover, the interviews showed that learners' chatting habits on social media might be consistent in chatbot-based learning, leading to high levels of concentration and excitement in fallacy learning for a long time. This state appears similar to a flow, an experience of deep engagement in an activity (Csikszentmihalyi, 2014). When experiencing a flow in learning, students concentrate on the learning tasks at hand, forget time-passing, and obtain high motivation and high learning efficiency (Csikszentmihalyi, 2014). Hence, chatbots may have mixed effects on fallacy learning from a behavioural aspect due to their integration with mobile social media applications.

Implications for researchers

So far as we know, no previous study has been conducted on chatbot-based learning of logical fallacies, the essential knowledge in EFL writing (El Khoiri & Widiati, 2017). This study contributed to this essential but understudied field by developing a chatbot-based learning programme of logical fallacies and exploring its perceived effectiveness, advantages and disadvantages in developing target fallacy knowledge and increasing motivation in fallacy learning. The findings indicated the effectiveness of chatbots in this area of EFL education. For example, this study identified the perceived usefulness of chatbots in consolidating target knowledge of logical fallacies in EFL writing by presenting exercises and immediate feedback, aligned with Ruan et al.'s (2019a) study of chatbot-based learning of EFL vocabulary and Hsu et al.'s (2021) study of chatbot-based learning of EFL speaking skills. Similarly, our interview results showed that chatbots could develop social connections with learners and thereby improve their affective states in fallacy learning, which was in line with Bailey et al.'s (2021) study on chatbot-based EFL learning. Based on the results, practitioners and researchers were recommended to apply chatbots in developing logical fallacies in EFL writing, taking more advantage of this powerful technology in the field of EFL education.

In addition, this study revealed new findings and tendencies rarely mentioned in previous studies on chatbot-based EFL education. Firstly, regarding the presentation of learning materials in chatbot-based EFL education, most previous studies focused on chatbot affordances of repeated knowledge delivery and different multimedia elements (e.g. Bailey et al., 2021; El Shazly, 2021). However, this study identified chatbots' affordance of the "one-at-a-time" presentation of learning materials and its mixed effects on fallacy learning. As shown in the interview and test results, chatbots' "one-at-a-time" presentation of learning materials may facilitate learners' acquisition and consolidation of target knowledge while reducing their efficiency in knowledge internalisation and elaboration. So far, little research has been conducted on the "one-at-a-time" presentation of learning materials in chatbot-based EFL learning. Considering the complex impacts of this chatbot affordance, we called for more studies in this direction.

Secondly, this study revealed the potential of integrating a self-regulated learning framework into chatbot-based learning of logical fallacies. The interview results showed that chatbots could encourage learners' application of planning and self-evaluation in fallacy learning, two essential self-regulated learning strategies (Zimmerman & Moylan, 2009). Our results also showed that chatbot-based learning was highly available and accessible and allowed high learner autonomy, which was conducive to self-regulated learning (Dent & Koenka, 2016). Considering the overall positive effects of self-regulated learning on EFL development (Zhang & Zou, 2022), it appears promising to enhance the effectiveness of chatbot-based learning of logical fallacies by integrating it into the self-regulated learning framework. So far, this promising direction has rarely been investigated, so future studies may work on it.

Thirdly, this study suggested that the experience of chatbot-based learning of logical fallacies appeared similar to a flow, the experience that usually results in high efficiency and positive emotions in learning (Csikszentmihalyi, 2014). Previous researchers had found that learners were likely to enter a flow when they had high autonomy and rich human-computer interactions (Perttula et al., 2017; Zou et al., 2021), which, as shown in our interview results, were affordable in chatbot-based learning of logical fallacies. Hence, chatbots may facilitate learners' entrance into a flow in fallacy learning, thereby leading to positive academic and affective outcomes. Since the integration of the flow theory into chatbot-based learning of logical fallacies remained limited, more research may be conducted on it in the future.

Finally, few previous studies compared chatbots with equivalent technology tools in EFL education (Huang et al., 2022). This study investigated chatbot-based learning of logical fallacies in EFL writing based on the comparison with website-based learning. Based on the comparison, we might have contributed to a more in-depth understanding of chatbot-based learning by identifying its advantageous features (i.e. high-quality human-computer interactions, study plan making and high accessibility) and mixedly-perceived features ("one-at-a-time" presentation of learning materials, conversational elements and learner concentration and perseverance) for developing logical fallacies in EFL writing. The findings echoed Huang et al.'s (2022) arguments concerning the value of comparing chatbots with equivalent technology tools in EFL education. Hence, future researchers may continue exploring chatbot-based EFL learning based on its comparison with other technology tools, such as robots and Personal Digital Assistants (PDAs).

Conclusions and limitations

This article presents an exploratory study of five-week chatbot-based learning of logical fallacies frequently seen in EFL writing. The results indicated that the chatbot was perceived as slightly less effective than the website in developing target knowledge of logical fallacies, while it was perceived as more effective than the website in improving learner motivation. Advantages of the chatbot over the website included high-quality human-computer interactions, supportiveness for making study plans and high accessibility. Learners have mixed perceptions of the "one-at-a-time" presentation of learning materials, conversational elements, and learner concentration and perseverance in chatbot-based learning. Based on the results, we explored how chatbots may influence fallacy learning based on the self-regulated learning framework. Future researchers may continue to work on chatbot-based learning of logical fallacies from the aspects of the "one-at-a-time" presentation of learning materials, the integration with the self-regulated learning framework and the flow theory, and the comparison against other technology tools.

There are several limitations in this study. First, our sample is small-sized, so we could not conduct inferential statistics. Future studies may investigate the effectiveness of chatbot-based learning with large sample groups, so the findings could be more generalisable based on inferential statistics. Second, this study focused on the effects of the chatbot on learner motivation and target knowledge of logical fallacies in EFL writing. Future studies may investigate its effects on other aspects, such as flow experiences in learning and learners' ability to reduce logical fallacies in authentic writing.

Finally, this study has revealed the chatbot's effectiveness in developing a new aspect of EFL knowledge, logical fallacies in EFL writing. Despite many advantages, chatbot-based learning is still of challenges, with complex features exerting mixed influences. Since this technology remains in its early stages, we call for more investigations on educational chatbots, such as the combination with other technology and the integration with learning theories. The application of ChatGPT in developing knowledge of logical fallacies also appears a promising direction.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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